

G. Lejeune Dirichlet's Works. II.

V.

**On the Reduction
of Positive Quadratic Forms
with Three Undetermined Integral Numbers.**

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**On
the Reduction of Positive Quadratic Forms
with Three Undetermined Integral Numbers.**

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It is well known that Lagrange first showed that every binary quadratic form can be reduced, that is, transformed into an equivalent form whose coefficients satisfy certain inequalities. At the same time he showed that in each class of positive forms there is always only one such form, so that, in this case, the several reduced forms belonging to a given determinant may serve as representatives of the several classes. After the ternary forms had later been treated from a general point of view in the *Disquisitiones arithmeticae*, the further development of this theory required the extension to positive ternary forms of the investigation carried out by Lagrange for the positive binary forms: one had to find inequalities among the coefficients such that in each class they are satisfied by one and only one form.

This extension, beset with great difficulties, was carried out by Seeber in a work devoted specifically to positive ternary forms. It forms the principal content of that work, which Gauss characterized as follows in a highly interesting review²:

Full justice must be done to the spirit of thoroughness with which these matters are treated. If we must regret that this thoroughness is accompanied by great length, perhaps discouraging to some—the solution of the problem taking 41 pages and the proof of the theorem 91 pages—this is not to be taken as a reproach. When a difficult problem or theorem is to be solved or proved, the first step, and one to be acknowledged with due thanks, is that a solution or proof be found at all. The question whether this could have been done in an easier and simpler manner remains idle until the possibility has also been decided by the deed. We therefore consider it premature to dwell here on this question.

The great complication of Seeber's method long ago tempted me to try to found the theory of reduced ternary forms in a simpler way. In now communicating the result of these efforts, I believe that, in the interest of brevity and, if I may so speak, of transparency of exposition, I must retain the geometrical form in which I have conducted the investigation. It is based on the remarkable relations which hold between quadratic forms with two or three elements and certain spatial configurations. I begin by carrying out the indications already given by Gauss, in the review just mentioned, concerning these relations.

¹An extract from this memoir had already been given in the Monthly Report of the Academy. There the principle of this new treatment of the reduction of positive ternary forms—the consideration of successive minima—was indicated, and the proof of the first of Seeber's two results was carried out completely according to this principle.

²Crelle's Journal, vol. 20, p. 312.

§. 1.

The ternary form

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy = \varphi, \quad (1)$$

in which we regard x, y, z as first, second, and third element, is called positive when φ never becomes negative for real values of these elements. In such a form the coefficients

$$a, \quad b, \quad c$$

are always positive, while the combinations of coefficients

$$a'^2 - bc, \quad b'^2 - ac, \quad c'^2 - ab, \quad aa'^2 + bb'^2 + cc'^2 - abc - 2a'b'c' = -D, \quad (2)$$

of which the last, $-D$, is called the determinant of the form, are negative³. By virtue of these conditions there are always three acute or obtuse angles λ, μ, ν , completely determined by the equations

$$\cos \lambda = \frac{a'}{\sqrt{bc}}, \quad \cos \mu = \frac{b'}{\sqrt{ac}}, \quad \cos \nu = \frac{c'}{\sqrt{ab}},$$

from which a trihedral angle can be formed, since the required condition

$$\cos^2 \lambda + \cos^2 \mu + \cos^2 \nu - 2 \cos \lambda \cos \mu \cos \nu < 1$$

coincides with $D > 0$.

Since, however, two mutually symmetric trihedral angles can be formed with the same angles λ, μ, ν , we agree always to choose that one of the two in which the edges, as they lie opposite these angles in order, follow one another from right to left with respect to a straight line drawn from the vertex 0 into the interior of the angle and thought of as going upward. If we now regard the three edges as the positive axes of a coordinate system, the whole infinite space can be referred to our form by taking the products $x\sqrt{a}, y\sqrt{b}, z\sqrt{c}$ as the coordinates of an arbitrary point. Then φ expresses the square of the distance of this point from the vertex, or, more generally, the square of the distance between two points whose like-named coordinates have those products as differences.

Now form, with three new undetermined elements x', y', z' , the linear expressions

$$x = \alpha x' + \beta y' + \gamma z', \quad y = \alpha' x' + \beta' y' + \gamma' z', \quad z = \alpha'' x' + \beta'' y' + \gamma'' z', \quad (3)$$

where the only restriction is that the determinant formed from the nine coefficients

$$\alpha\beta'\gamma'' + \beta\gamma'\alpha'' + \gamma\alpha'\beta'' - \gamma\beta'\alpha'' - \alpha\gamma'\beta'' - \beta\alpha'\gamma'' = E \quad (4)$$

be not zero. Then φ goes over into a new form φ' , and the corresponding quantities for that form will be denoted by accented letters⁴.

If one again lets an infinite space correspond to the new form, the two infinite spaces are thereby put in pointwise correspondence: two points correspond when, in the expressions for their coordinates

$$x\sqrt{a}, y\sqrt{b}, z\sqrt{c}; \quad x'\sqrt{a'}, y'\sqrt{b'}, z'\sqrt{c'},$$

the elements x, y, z and x', y', z' are connected by equations (3). If these expressions are the coordinate differences for two pairs of corresponding points, then the same relations among $x, y, \dots$ hold as before; hence, by what has just been said and because $\varphi = \varphi'$, the distance

³*Disquisitiones arithmeticae*, art. 271.

⁴This is the meaning determined here for the accented letters in this paragraph; it is therefore essentially different from the meaning of a', b', c' in the first display of §. 1. K.

between any two points of the one space is equal to the distance between the corresponding two points of the other. The two pointwise corresponding spaces are therefore either congruent or symmetric: when the origins 0 and $0'$ are made to coincide, either every point falls on its corresponding point or on the opposite point of the latter, where, for brevity, two points of the same space which lie at equal distance and in opposite directions from the origin are called opposite points.

To decide which of these two cases occurs, one draws from the vertex in one space to three arbitrary points and examines whether the straight lines drawn in the other space from its vertex to the corresponding points have the same or the opposite order. For example, in the second space take the lines to the points with coordinates

$$\sqrt{a'}, 0, 0; \quad 0, \sqrt{b'}, 0; \quad 0, 0, \sqrt{c'},$$

lying on the positive axes. By the convention above these follow one another from right to left. For the corresponding points in the first space the coordinates are

$$\alpha\sqrt{a'}, \alpha'\sqrt{a'}, \alpha''\sqrt{a'}; \quad \beta\sqrt{b'}, \beta'\sqrt{b'}, \beta''\sqrt{b'}; \quad \gamma\sqrt{c'}, \gamma'\sqrt{c'}, \gamma''\sqrt{c'}.$$

Their order is the same or the opposite according as the determinant E is positive or negative. Thus the transformation is a motion when $E > 0$ and a motion followed by reflection when $E < 0$.

If the elements x, y, z had arbitrary values up to now, let them henceforth be restricted to integers. In place of the whole space we then have an infinite system of parallelepipedally arranged points, that is, a system formed by the intersections of three families of parallel equidistant planes. Suppose now that the substitution coefficients $\alpha, \beta, \gamma, \dots$ are also integers and that E has the value ± 1 . Then to each integral combination x, y, z there corresponds an integral combination x', y', z' , and conversely. By what has just been shown, the parallelepipedal systems so put in correspondence can be brought into such a position that one coincides either with the other or with the opposite points of the other. Since, however, the opposite points of such a system again form the same system, the two cases are not distinct. This is also clear from the fact that φ' is unchanged when one changes the signs of $\alpha, \beta, \gamma, \dots$, which changes E into $-E$.

Thus the two systems are always congruent. Systems corresponding to two equivalent ternary forms φ and φ' are therefore the same spatial configuration in two different arrangements. Conversely, any two different parallelepipedal arrangements of the same system correspond to equivalent forms. Indeed, choose any point of the system as common origin. Between the coordinates relative to the two systems of axes, and hence also between the proportional elements x, y, z, x', y', z' , there are linear equations without constant term, that is, equations of the form (3). Since, by assumption, integral values of x, y, z are accompanied by integral values of x', y', z' and conversely, the coefficients $\alpha, \beta, \gamma, \dots$ are also integers and $E = \pm 1$. On the other hand, for corresponding integral values of the elements one has $\varphi = \varphi'$, and the equality therefore holds identically.

Analogous relations hold between a positive binary form

$$lx^2 + 2mxy + ny^2$$

and a system of parallelogrammatically arranged points. Here take two axes inclined to one another by the angle ϑ determined by

$$m = \sqrt{ln} \cos \vartheta,$$

where the distinction between the axes is made uniformly, for instance by choosing the second to the left of the first after one side of the plane has been designated as the upper side. If $x\sqrt{l}, y\sqrt{n}$

are then regarded as coordinates, one obtains a system of points completely determined by the quadratic form, which may be regarded as the intersections of two families of equidistant parallel lines. If between two forms there is the so-called proper equivalence, so that $\alpha\delta - \beta\gamma$ in the substitutions $x = \alpha x' + \beta y'$, $y = \gamma x' + \delta y'$ equals positive unity, then the corresponding systems may be brought to coincide by a motion in the plane. In the other case, where $\alpha\delta - \beta\gamma = -1$, generally speaking one of the systems must be turned over for this purpose.

§. 2.

Having established the connection between quadratic forms and certain geometrical configurations, we must develop some further properties of these configurations. For brevity, a system of parallelogrammatically or parallelepipedally arranged points will be called a *system of second* or *third order*, while an infinite row of equidistant points in a straight line will be called a *system of first order*.

The common character of all three kinds of systems is plainly this: if such a system is brought by a motion without rotation, which we shall call a displacement, into a new position such that one of its points passes into the place originally occupied by another, then the same happens for all points; the system in its new position coincides completely with the system in its original position. It is easy to prove that this displacement property completely characterizes all three kinds of systems: a system with this property is respectively a system of first, second, or third order according as it lies in a straight line and contains two points, lies in a plane and contains three non-collinear points, or contains at least four points not lying in one plane.

For example, suppose all points of a system lie on the same straight line, and let a and a' be two neighbouring points of it. By a displacement carrying a to a' , the point a' goes to a point a'' which is as far from a' as a' is from a . Thus a'' also belongs to the system, and the system has no point between a' and a'' , since such a point would before the motion have lain between a and a' . Continuing this consideration indefinitely in both directions proves the assertion.

Now suppose a planar system has the displacement property, and let a and a' be two neighbouring points such that no point of the system lies between them on the line aa' . Since the displacement from a to a' carries the infinite line aa' into itself, all points of the system on this line form a system of first order $\dots a''aa'a'' \dots$. Since, by hypothesis, the system has at least one further point outside this line, let b be one of the points nearest to the line. If now a displacement carries a to b , the first-order system passes into the new position $\dots b''bb'b'' \dots$ and in that position belongs to the original system. It is also clear that neither between the points $\dots b'', b, b', b'', \dots$ nor between the lines $\dots bbb' \dots, \dots aaa' \dots$ can any point of the system lie. Continuing in this way, one sees that the whole system can be arranged parallelogrammatically, and that $aa'bb'$ may be chosen as one of its fundamental parallelograms. This construction also gives all parallelogrammatic arrangements of which the system is capable: a' may be chosen arbitrarily subject only to the necessary restriction that no point lie between a and a' , and then b may be chosen arbitrarily in the nearest parallel line.

Finally, suppose a system with the displacement property contains at least four points not lying in the same plane. Through any three non-collinear points of it draw a plane. Since every displacement parallel to this plane moves the plane into itself, the points lying in it form, by the preceding result, a system of second order. After this planar system has been divided parallelogrammatically in any way, take one of the remaining points of the spatial system nearest to the plane and displace the system so that an arbitrary point in the plane goes to the chosen point outside it. Repeated application of this motion and of its inverse evidently gives a parallelepipedal arrangement of the given system. At the same time the construction plainly has the required generality, since the first plane, the arrangement of the system of second order in it, and finally the choice of the point in the nearest plane may all be made arbitrarily.

At the end of this paragraph we also show that however one divides the same system of second or third order, the parallelogram or parallelepiped underlying the division always has the same content. This is the geometrical meaning of the theorem that equivalent forms have equal determinants. Namely, in the plane of a system of second order think of a closed curve, for instance a circle. Let z denote the area enclosed by it and s the number of points inside it; it is immaterial whether points on the circumference are counted or not. The quotient z/s , as the radius increases, clearly has the content of a fundamental parallelogram as its limit. Since s and z are independent of the kind of arrangement of the system, the theorem follows for systems of second order. The assertion for spatial systems follows in exactly the same way.

§. 3.

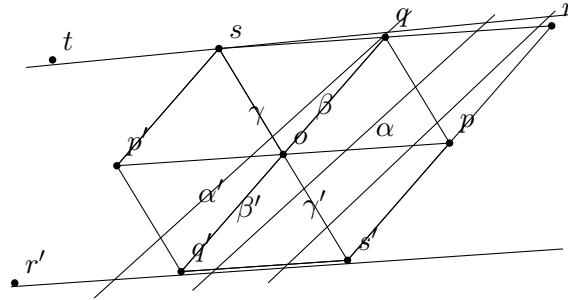
We now show that every system of second order can be divided according to a fundamental parallelogram whose sides are not greater than its diagonals.

I. Let o be any point of the system. The remaining points occur in pairs equidistant from o and in opposite directions. Let p be one of the pair for which the distance from o is smaller than for every other pair; if this shortest distance occurs for more than one pair, choose one of them arbitrarily.

The given system consists of infinitely many mutually congruent and equidistant systems of first order, one of which contains o and p . In one of the two adjacent systems take the point q nearest to o , or, if the same shortest distance occurs for two points, either one. The parallelogram $poqr$ so obtained has the required properties, since by construction

$$op \leq oq, \quad oq \leq or, \quad oq \leq os = pq.$$

A fundamental parallelogram satisfying these conditions will henceforth be called *reduced*.



II. We now establish the relation between such a parallelogram and the planar system to which it belongs. If $poqr$ is reduced, then without loss of generality the angle poq may be assumed not obtuse; if it were obtuse, the adjacent parallelogram belonging to the same arrangement would have an acute angle at o . Similarly we may assume $op \leq oq$. Then $or > oq$, and it remains only to consider the condition $pq \geq oq$. Put $op = \sqrt{l}$, $oq = \sqrt{n}$, so that $l \leq n$. Then the relation to the whole system is this: the minimum distance of any point of the system from o is $\sqrt{l}$, and after a point at this distance has been chosen, the second minimum, in all remaining directions outside the line through o and that first point, is $\sqrt{n}$.

This statement is general. The further assertion that the first minimum is attained only at p (opposite points being counted only once), and the second only at q , has the following exceptions:

- 1°. If $op < oq$ and $oq = pq = os$, the first minimum occurs only at p , the second at q and s .
- 2°. If $op = oq$ and $oq < pq = os$, the minima are equal, and p and q may be interchanged.

3°. If $op = oq = pq = os$, one of p, q, s may be chosen as first point, and then either of the remaining two as second.

To prove this it is enough, since opposite points are always equally distant from o , to show that q is nearer to o first than all other points on the line sqr , except s , whose distance from o equals $os = pq \geq oq$, and secondly than all points on the following parallel lines.

Since $pq \geq op$, $pq \geq oq$, and the angle poq is not obtuse, the triangle opq , and hence also the congruent triangle oqs , has no obtuse angle. Thus the perpendicular from o to qs falls between s and q (inclusive), proving the first point. Write

$$\cos poq = \frac{m}{\sqrt{ln}},$$

with $m \geq 0$. Then

$$pq^2 = l - 2m + n \geq oq^2 = n,$$

and hence

$$2m \leq l, \quad 2m \leq n, \quad 4m^2 \leq ln.$$

If the square of the height of the parallelogram (with $op = \sqrt{l}$ as base) is k , then the square Δ of its area is

$$\Delta = lk = ln - m^2 \geq \frac{3}{4}ln,$$

so that

$$\sqrt{k} \geq \frac{1}{2}\sqrt{3n}.$$

Thus already the second parallel line is at least $\sqrt{3n} = oq\sqrt{3}$ away, proving the second point.

III. Since the successive minima $\sqrt{l}$, $\sqrt{n}$ are determined by the system itself and independently of any particular arrangement, while by what has just been shown they agree in magnitude with the sides of a reduced parallelogram, it follows that if the system admits several such arrangements, the sides of the reduced parallelograms always keep the values $\sqrt{l}$ and $\sqrt{n}$.

All possible fundamental parallelograms are therefore obtained by drawing from o lines to all nearest points (again excluding opposite points), and then taking, in the corresponding nearest parallel line, the nearest point or the two nearest points. By II, these nearest points are nearer to o than all points of the following parallel lines; hence one may omit the separate requirement that the second points be in the first parallel line. Thus all possible arrangements of the system arise by connecting o successively with all pairs of points for which the successive minima occur. Taking account of II, there is in general, and in the second singular case mentioned there, only one arrangement; in the first and third exceptional cases there are respectively two and three.

In the present notation those singular cases correspond to

$$2m = l < n, \quad 2m < l = n, \quad 2m = l = n.$$

§. 4.

We have so far considered only properties of the geometrical configurations which may be regarded as constructive representations of known results from the theory of forms. We now solve a different problem: given a system of second order and a fixed point o of it, determine the part of the plane in which every point is nearer to o than to any other point of the system. Since the condition that a point is not farther from o than from another point v is that it lie on the same side as o of the perpendicular erected at the midpoint of ov , we must combine o with all other points and construct the convex polygon formed by the corresponding perpendiculars. Of these infinitely many perpendiculars only finitely many matter; the others do not meet the polygon

determined by these. We keep the previous assumptions, so that in the reduced parallelogram ($poqr$) one has $op \leq oq$, the angle poq is not obtuse, and the angles opq , oqp are acute.

Under these assumptions, one easily proves that only the six vertices p, q, s, p', q', s' of the four parallelograms meeting at o need be considered, and that even the perpendiculars corresponding to s and s' merely touch the desired figure in the special case where poq is a right angle; in that case the same is true of those corresponding to r and r' .

Draw the lines $pq, os, p'q', os'$. One obtains the congruent triangles

$$poq, \quad qos, \quad sop', \quad p'oq, \quad q'os, \quad s'op.$$

If only p, q, s, p', q', s' are considered, one erects perpendiculars at the midpoints of the lines from o to those points, that is, one performs the same construction as if one wanted the circumcentres of the triangles just listed. Since no angle in these triangles is obtuse, two consecutive perpendiculars do not meet outside the corresponding triangle. One obtains the hexagon $\alpha\beta\gamma\alpha'\beta'\gamma'$, with centre o and equal opposite angles and sides, as the region in which every point is nearer to o than to any one of the points p, q, s, p', q', s' . One readily verifies that, except for r and r' , the perpendiculars corresponding to the remaining points do not meet this hexagon. By symmetry it suffices to prove this for points in and above the line pop' . For points on the line this is clear; for the points above it, it follows because their distance from o is greater than the diameter of the circle circumscribed about the hexagon. If ϱ denotes the square of its radius, then

$$4\varrho\Delta = \ln(l - 2m + n),$$

and from $2m \leq l$, $2m \leq n$, $\Delta \geq \frac{3}{4}\ln$ follows

$$4\varrho \leq \frac{4}{3}(l - 2m + n) \leq \frac{8}{3}n.$$

Since the square of the distance from o of points of the second and following parallel lines is at least $3n$, it remains only to consider the points in $tsqr$ other than s, q, r . None of these is nearer to o than t , for which the square of the distance is $4l - 4m + n$. That this is greater than 4ϱ is seen at once after multiplication by Δ and use of $2m \leq l \leq n^5$. For the point r one likewise sees that the square of its distance from o is $l + 2m + n > 4\varrho$, except in the single case $m = 0$, where the corresponding perpendicular only touches.

It is thus proved that a point lies inside the hexagon $\alpha\beta\gamma\alpha'\beta'\gamma'$, and only then, precisely when it is nearer to o than to any other point of the system. On each side, the distance from o becomes equal to the distance from a second point—for example, for $\alpha\beta$ this point is q —and each vertex of the figure is equally distant from o and from two further points of the system. This last statement changes only in the special case where poq is a right angle; then β and γ , and likewise β' and γ' , coincide, and the hexagon becomes a rectangle whose vertices are equidistant from o and from three further points of the system.

It is clear that the same hexagon is obtained whichever reduced parallelogram is used in the singular cases where more than one exists. Likewise, the hexagons or quadrilaterals corresponding to all points of the system are congruent and cover the whole plane.

We also note that the expression

$$\varrho = \frac{\ln(l - 2m + n)}{4(\ln - m^2)}$$

decreases when, with l and n held constant, m increases from zero to its limiting value $\frac{1}{2}l$. Hence

$$\varrho \leq \frac{1}{4}(l + n) \leq \frac{1}{2}n. \quad (5)$$

⁵The difference $4l - 4m + n - 4\varrho$, multiplied by Δ , can be written as $4m^3 + n(l^2 - m^2) + \ln(l - 2m) + l(\ln - 4m^2)$, in which the first two terms are positive and the last two non-negative. K.

There is also the inequality

$$2\Delta(n - \varrho) \geq ln^2, \quad (6)$$

whose truth follows immediately after multiplying by 2, bringing all terms to one side, and substituting $\Delta = ln - m^2$, $4\Delta\varrho = ln(l - 2m + n)$, whereby it becomes

$$ln(l - 2m) + 2mn(n - l) \geq 0.$$

§. 5.

We now come to the proper subject. We must show that every system of third order can be arranged according to a parallelepiped whose faces are reduced parallelograms, and whose edges, four by four equal, do not exceed its diagonals.

Having fixed an arbitrary point (0) of the system, choose in the pair of opposite points for which the distance from (0) is a minimum, or if this minimum occurs for several pairs, arbitrarily in one of these pairs, a point (1).

Among all points outside the line (01) choose again one of the two nearest, call it (2), with arbitrary choice if the same shortest distance occurs for several pairs. Since in the whole system, apart from points on (01), no point is nearer to (0) than (2), the same holds in the plane (102); for the system contained in that plane, (102) is a reduced parallelogram (§. 3, III.). Now take in one of the two nearest parallel planes the point nearest to (0), or, if the minimum occurs more than once, one of the nearest, and join (0) to the chosen point (3). The parallelepiped with edges (01), (02), (03) will then satisfy the requirement. First, by construction,

$$(01) \leq (02) \leq (03).$$

For the base faces of the parallelepiped—that is, the opposite faces containing the two edges that do not exceed the third in size; the other four faces will be called side faces—we already know that they are reduced. By the double inequality just noted we have only to show that the four diagonals of the side faces and the four body diagonals are not smaller than (03). These eight diagonals agree in length with the eight lines from (0) to the eight points lying around (3) in the plane of the upper base face, the eight vertices of the four parallelograms meeting at (3). None of these connecting lines is smaller than (03), by the condition under which (3) was chosen.

Having seen that a system of third order can always be divided according to a reduced parallelepiped, we now establish the relation between such a parallelepiped and the system, and especially compare the distances of the points of the system from (0). Put

$$(01) = \sqrt{a}, \quad (02) = \sqrt{b}, \quad (03) = \sqrt{c},$$

and keep throughout the assumption $a \leq b \leq c$.

- 1°. In the plane of the base face, the relations discussed above (§. 3, II.) hold; hence the successive minima of distance have magnitudes $\sqrt{a}$, $\sqrt{b}$, with the possible freedom in the singular cases described there.
- 2°. Consider now the points outside the plane of the lower base face, first those in the plane of the upper base face. Since, by the assumption that the parallelepiped is reduced, the line (03) is not greater than any line drawn from (0) to the eight points lying around (3), the foot of the perpendicular from (0) to the plane of the upper base face is not farther from (3) than from any of those eight points. Thus this foot does not fall outside the hexagon or quadrilateral belonging to (3) that was constructed in the preceding paragraph. Among the eight points, exceptionally, if the foot falls on a side, one point, or if the foot coincides

with a vertex, two points (three when the polygon becomes a rectangle), may be as near to the foot as (3) is; all other points of the plane are farther away from it. It follows that the shortest distance from (0) to a point of the upper base face, equal to $\sqrt{c}$, occurs generally only for (3), but exceptionally may occur for one, two, or even three further points.

- 3°. For the following parallel planes we need a bound for the square h of the perpendicular just mentioned. Since its foot does not fall outside the hexagon belonging to (3), if ϱ denotes the square of the radius of the circumscribed circle, then

$$h \geq c - \varrho.$$

But by §. 4 also $\varrho \leq \frac{1}{2}b \leq \frac{1}{2}c$, hence $h \geq \frac{1}{2}c$. Thus already the second parallel plane is at least $\sqrt{2c}$ distant, so above the upper base face there are only points whose distance from (0) is greater than $\sqrt{c}$.

Summing up, the minimum distance in the whole system has value $\sqrt{a}$; after a point at this distance has been chosen, the minimum in the remaining directions is $\sqrt{b}$; and after the second point has also been fixed, the least distance from (0) to points outside the plane determined by (0) and the two first points is $\sqrt{c}$. The successive minima $\sqrt{a}$, $\sqrt{b}$, $\sqrt{c}$ are therefore completely determined in magnitude, though not always in position. The possible exceptions are exactly those arising from the singular cases in the base plane, together with the exceptional positions in the upper base plane described above.

Consequently every reduced arrangement of the system arises by choosing, in succession, points at which these successive minima are attained. In the ordinary case the arrangement is unique. If equalities among the minima or among the boundary conditions occur, several reduced parallelepipeds may appear; but they are precisely those obtained by the stated alternative choices of the points realizing the successive minima.

It follows in particular that if a reduced parallelepiped is such that every segment which is required not to exceed certain others is in fact strictly smaller than them, then the corresponding system has no second reduced arrangement. Conversely, a second reduced arrangement can occur only when one of the limiting equalities holds.

Thus, when a system admits more than one reduced parallelepiped, knowledge of one arrangement suffices to find all the others. The first, non-singular case occurs always and only when the reduced parallelepiped given by this arrangement is of such a nature that all the lines which must not be exceeded by others are in fact strictly smaller: all face diagonals are greater than the corresponding sides, and all body diagonals of the parallelepiped are greater than its edges.

§. 6.

We now translate the results of the preceding paragraph to ternary forms. For uniformity, and to avoid useless distinctions, suppose that by interchanging or changing the signs of the undetermined elements—which does not remove the form from its class—each ternary form

$$ax^2 + by^2 + cz^2 + 2a'yz + 2b'xz + 2c'xy \tag{7}$$

has first been put into a shape such that $a \leq b \leq c$; secondly, among the coefficients a', b', c' , unless all three are non-zero and negative, none has the negative sign; and thirdly, if $b = c$, then c' , disregarding its sign, is not greater than b' , if $a = b$, then b' is not greater than a' , and if $a = b = c$, then neither c' is greater than b' nor b' greater than a' . It is easy to see that these requirements can be satisfied in only one way. Their introduction has the advantage that, just as before every ternary form corresponded to a completely determined parallelepiped, now to every

parallelepiped there belongs an analytic expression whose coefficients are completely determined also in their order and in their signs.

Under this convention we call the form (1), for which $a \leq b \leq c$, *reduced* when it corresponds to a reduced parallelepiped. Since the diagonals of the faces must not be smaller than the sides of those faces, one has

$$a \pm 2c' + b \geq b, \quad a \pm 2b' + c \geq c, \quad b \pm 2a' + c \geq c.$$

Put $\sigma = -1$ when a', b', c' are all three negative, and otherwise $\sigma = 1$. These conditions are equivalent to

$$a \geq 2c'\sigma, \quad a \geq 2b'\sigma, \quad b \geq 2a'\sigma, \quad (8)$$

and only when equality occurs will one of the diagonals in the corresponding parallelogram be equal to a side. The conditions concerning the body diagonals give

$$a + b + c + 2a'\delta + 2b'\varepsilon + 2c'\delta\varepsilon \geq c \quad (\delta = \pm 1, \varepsilon = \pm 1),$$

where the signs in $\delta = \pm 1, \varepsilon = \pm 1$ are arbitrary.

First take the case in which none of a', b', c' is negative. Considering the four sign combinations and the condition that, if a and b are equal, then $b' \leq a'$, one sees at once that the preceding inequality is then automatically satisfied by virtue of the conditions already obtained. The limiting case of equality, in which a body diagonal becomes equal to the edge $\sqrt{c}$, can occur only once, and only when one of b', c' is zero and when, at the same time, among the conditions (2), both the one involving the other of the two quantities b', c' and the one involving a' are limiting equalities. If a', b', c' are negative, the inequality is always satisfied, and without a limiting equality except when $\delta = \varepsilon = 1$. Thus one must add the new condition

$$a + b + 2a' + 2b' + 2c' \geq 0, \quad (9)$$

where again the lower sign refers to a diagonal becoming equal to the edge $\sqrt{c}$.

If the conditions (2), and also (3) when a', b', c' are negative, are satisfied without equality in any of the inequalities, then in the class to which the form belongs there is no second distinct reduced form, whether with or without equality in the defining conditions. This follows from the end of the preceding paragraph: the corresponding point system can then be divided according to only one reduced parallelepiped. The situation is different when not all conditions hold with strict inequality. Then several reduced forms may occur in the same class, and they can be derived from a given one. It suffices to show this in a principal case; we choose the case $b < c$.

Assume first $a > 2c'\sigma$. Then only the direction of the edge $\sqrt{c}$ can be changed, namely if in the plane of the upper base face there are one or more further points whose distance from the vertex is $\sqrt{c}$. If $\xi, \eta, 1$ are the values of the elements corresponding to such a point, then, when the third edge is directed toward it, all coefficients remain unchanged except a', b' , which become respectively

$$a' + c'\xi + b\eta, \quad b' + a\xi + c'\eta,$$

as is seen easily and almost without calculation.

By the preceding enumeration, the values of ξ, η satisfying the condition are, when $a = 2b'\sigma$,

$$\xi = -\sigma, \quad \eta = 0;$$

when $b = 2a'\sigma$,

$$\xi = 0, \quad \eta = -\sigma;$$

when simultaneously $a = 2b', b = 2a', c' = 0$,

$$\xi = -1, \quad \eta = -1;$$

and when a', b', c' are negative and satisfy

$$a + b + 2a' + 2b' + 2c' = 0,$$

then

$$\xi = 1, \quad \eta = 1.$$

Corresponding to these four hypotheses, a', b' are transformed into

$$a' - c'\sigma, b' - a\sigma(= -b'); \quad -a', b' - c'\sigma; \quad -a', -b'; \quad a' + b + c', a + b' + c'.$$

The third case, and in general the assumption $c' = 0$, may be disregarded, since it gives a new form which, after the prescribed changes of sign at the beginning of this paragraph, is plainly identical with the original one. In each of the three remaining cases one obtains, after making the necessary sign changes, a new reduced form in the same class, unless it coincides with the original. If two of the hypotheses hold simultaneously, two such forms are obtained. This completes the enumeration, since all three hypotheses clearly cannot hold simultaneously. If, still under the assumption $b < c$, one had $a = 2c'\sigma$, then, if $a < b$, the second edge in the base face would have had to be turned; if $a = b$, the first edge could also have been brought into the original position of the second; and this new position, or these two new positions, of the first two edges would have had to be combined with all directions of the third, including the original one.

The inconvenience that in singular cases several reduced forms may occur in the same class can easily be removed by adding to the general definition certain secondary conditions for such cases. They arise, for example, if one requires that the last coefficient c' , when it is not completely determined, have the smallest numerical value of which it is capable among the reduced forms of the class, and then proceeds similarly with b' . To illustrate this, consider among the singular cases just treated the case $b < c$, where none of the three conditions (2) holds with the lower sign, but the three negative

values a', b', c' satisfy

$$a + b + 2a' + 2b' + 2c' = 0.$$

As noted above, c' is then determined, and there are only two reduced forms. If a' and b' are the values of the fourth and fifth coefficients in one of them, then in the other they are $a' + b + c'$, $a + b' + c'$, or, since these latter values are evidently positive while c' is negative and therefore z must be replaced by $-z$ to satisfy the sign convention, rather

$$-(a' + b + c'), \quad -(a + b' + c').$$

As is natural, substituting these values for a', b' again gives the equation

$$a + b + 2a' + 2b' + 2c' = 0,$$

and the original values a', b' arise from them in the same way as they arose from a', b' . Thus the fifth coefficient can have only the two negative values b' and $-(a + b' + c')$, whose sum is $-a - c'$. Hence, if one adds to the defining conditions the further condition

$$-b' \leq \frac{1}{2}(a + c'),$$

the class will contain only one reduced form.

To conclude the memoir we derive from our principles a beautiful theorem found by Seeber by induction and proved by Gauss in the often mentioned review. According to this theorem, in a reduced form the product of the first three coefficients is not greater than twice the absolute value of the determinant.

Since the absolute value of the determinant is equal to the square of the volume of the parallelepiped corresponding to the form, the inequality to be proved, in the notation used in §. 5, 3°, is

$$abc \leq 2\Delta h,$$

where Δ denotes the square of the base area. Put

$$c = b + t,$$

where $t \geq 0$. From the inequality obtained in §. 5, 3°,

$$h \geq c - \varrho = b - \varrho + t,$$

after multiplying it by 2Δ , subtract the equation

$$abc = ab^2 + abt.$$

This gives

$$2\Delta h - abc \geq 2\Delta(b - \varrho) - ab^2 + (2\Delta - ab)t.$$

But by the inequality proved at the end of §. 4, $2\Delta(b - \varrho) - ab^2$ is non-negative, and $2\Delta - ab \geq \frac{1}{2}ab$ is positive. The theorem follows.